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Open-Source Embedding Models: A Comprehensive Survey of Techniques, Benchmarks, and Applications

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ABSTRACT Embedding models are fundamental to modern Natural Language Processing (NLP), transforming textual data into dense vector representations that capture semantic meaning and relationships. This systematic survey focuses exclusively on the landscape of open-source embedding models, tracing their evolution from early word embeddings (Word2Vec, GloVe) to advanced contextual and multimodal architectures (BERT, Sentence-BERT, BGE, Nomic Embed). Moving beyond descriptive narratives, our core contribution is a structured, comparative analysis of these models' operational tradeoffs, including model size, embedding dimensionality, inference latency, and memory consumption. This systematic review spans the literature up to 2025 drawing upon 70 papers to provide a critical assessment of standardized benchmarks, such as MTEB, and their limitations regarding bias and robustness. Finally, the paper concludes by discussing the broad applicability of embedding models across diverse domains, including search engines, recommendation systems, fraud detection, and healthcare, and offer recommendations for effective model selection and utilization.

INDEX TERMS Open source, Natural Language Processing (NLP), Embedding models, multimodal architectures, Retrieval-Augmented Generation (RAG) systems.

I. INTRODUCTION

EMBEDDING models have become the cornerstone of modern Natural Language Processing (NLP), serving as a fundamental layer for a wide array of downstream tasks [1]. These models function by transforming textual data, whether individual words, short phrases, complete sentences, or entire documents, into dense numerical vectors within a high-dimensional space [2](see Figure 1).

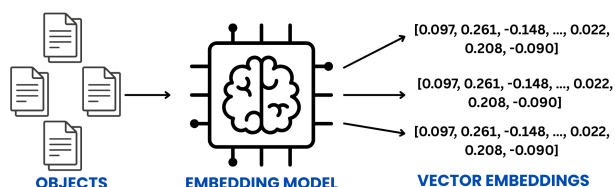


FIGURE 1: Illustration of how textual objects are converted into vector embeddings.

The key principle behind this transformation is to capture semantic meaning and the intricate relationships that exist

between different textual units [1]. By representing language in this numerical format, embedding models enable machine learning algorithms to efficiently process and understand the nuances of human language [3].

The significance of embedding models lies in their ability to convert qualitative textual data into a quantitative form that machines can readily interpret and manipulate [?], [4]. This numerical representation allows the application of mathematical and statistical techniques to analyze and compare textual information, which is crucial for tasks such as semantic search, personalized recommendation, large-scale information retrieval, text classification and clustering, and question answering systems [5], [6]. The ability to represent the meaning of text as numerical vectors has revolutionized how machines interact with and understand human language.

The evolution of embedding models reflects a significant paradigm shift in the field of NLP. Early approaches relied on symbolic representations, treating words as discrete, unrelated entities. The advent of word embeddings marked a turning point, allowing semantic similarities to be cap-

tured in a continuous vector space [1]. These embeddings enabled algebraic operations on meaning, such as analogies and concept relatedness, moving beyond keyword matching toward deeper semantic understanding. Classical models like Word2Vec, GloVe, and FastText [7]–[9] exemplify this era.

As NLP tasks grew more complex, there was a need to represent larger textual units. This gave rise to sentence embeddings, where models like Sentence-BERT [10], Universal Sentence Encoder [11], and MPNet [12], [13] encapsulate the overall meaning of sentences or paragraphs. More recently, multimodal embeddings have emerged, integrating textual and non-textual data (e.g., images, audio, graphs) into unified representations. Open-source models such as Nomic Embed Vision [14], [15] and OpenCLIP [16] demonstrate this expansion of scope.

While these evolutionary milestones laid the groundwork for today's embedding landscape, the field is now undergoing a critical paradigm shift driven by efficiency and accessibility. Previous surveys in the literature predominantly focus on word embeddings or earlier, primitive forms of representation learning, and often emphasize benchmark performance in isolation. In contrast, this work provides a systematic and forward-looking review of open-source embedding models (< 2025). To the best of our knowledge, this is the first comprehensive survey dedicated exclusively to modern open-source embedding models. Beyond descriptive performance reporting, we conduct a structured comparative meta-analysis of critical operational trade-offs, including model size, embedding dimensionality, memory footprint, and inference latency. This approach provides practitioners with the crucial, resource-aware guidance needed for effective deployment, positioning this manuscript as a vital resource for both synthesizing recent developments and shaping the future direction of open-source, efficient embedding research.

A. DATA MODALITY

Embedding models today exhibit versatility across multiple data modalities [17]. Text embeddings remain foundational, spanning word-level [18]–[20], sentence-level [11], [21], and document-level representations. Image embeddings, pioneered by models like CLIP and its open-source counterpart OpenCLIP, leverage contrastive learning to align images and text. In audio, models such as Wav2Vec 2.0 [22] and HuBERT [23] generate robust speech embeddings, enabling tasks like classification, speaker identification, and audio-text retrieval (e.g., CLAP and AudioCLIP). Furthermore, graph embeddings extend representation learning to relational and structural data. Frameworks like PyTorch Geometric and DGL support GNN-based models such as GraphSAGE [24], [25] and Graph Attention Networks (GATs) [26], with applications in social networks, knowledge graphs, and bioinformatics.

As shown in Figure 2, sentence embeddings currently occupy the largest share (35%), followed by word embeddings (30%), multimodal embeddings (20%), and document embeddings (15%). These values are indicative estimates,

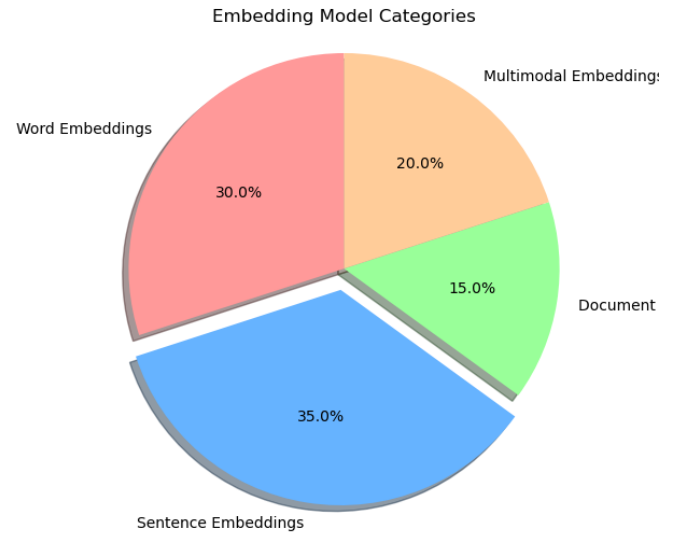


FIGURE 2: A pie chart showing the categories of embedding models

informed by recent industry and academic researches: for example, sentence embeddings align with Anthropic's 2025 Economic Index for RAG tasks (36%) [27], word embeddings with Stack Overflow usage statistics (28%) [28], multimodal embeddings with McKinsey 2025 reports on agentic and multi-modal scaling (23%) [29], and document embeddings based on a literature review of specialized embedding models (~10–15%). While not exhaustive, this snapshot reflects the relative prominence of each category and underscores the growing importance of sentence and multimodal embeddings in modern NLP. A more precise quantification would require systematic analyses of model adoption and real-world deployment trends across tasks and languages.

B. EVALUATION AND BENCHMARKS

To evaluate and compare embedding models, the NLP community relies on standardized benchmarks. Platforms such as the Massive Text Embedding Benchmark (MTEB) [30] provide a comprehensive suite of tasks, including information retrieval, clustering, classification, and semantic textual similarity, spanning multiple languages and domains. Public leaderboards, notably those hosted by Hugging Face [31], track ongoing progress and surface state-of-the-art embedding models across these tasks. It is important to note that the benchmark data and leaderboard results referenced in this study reflect a snapshot taken on **April 15, 2025**. Given the rapid pace of development in representation learning, model rankings and reported performance may evolve significantly beyond this point in time. Despite their utility, benchmark results must be interpreted with caution, as strong performance on standardized tasks does not always translate directly

to real-world applications [6]. Beyond aggregate accuracy metrics, factors such as representational bias, embedding dimensionality, robustness to perturbations, and sensitivity to contextual variation remain central evaluation challenges. Furthermore, benchmark outcomes often reveal trade-offs between open-source and proprietary embedding models, influencing practical considerations around deployment, cost, transparency, and long-term adoption.

C. SURVEY METHODOLOGY AND SCOPE

This manuscript is designed as a systematic, transparent, and reproducible survey of open-source embedding models. The scope is explicitly bounded to non-proprietary models and publicly available scientific literature published or substantially updated between 2013, marking the introduction of modern vector embeddings with Word2Vec—and early 2025. The survey traces the evolution from static word embeddings to contextual, multimodal, and efficiency-oriented architectures (e.g., BERT, SBERT, BGE, and Nomic).

Models included in this review must (i) be fully open-source, with open model weights; (ii) demonstrate relevance to general NLP or multimodal retrieval tasks; and (iii) report empirical evaluations on widely adopted public benchmarks, particularly MTEB or MMTEB, along with operational characteristics such as model size, embedding dimensionality, or inference latency. Proprietary or closed-source APIs are excluded from in-depth technical analysis and are referenced only for historical context or high-level comparison. Models limited to highly specialized or narrow application domains without broad research applicability are also excluded. To ensure objectivity and reproducibility, a structured systematic review workflow was followed. The guiding research question is: How have open-source embedding models evolved from static word vectors to contextual and multimodal architectures, and what operational tradeoffs govern their deployment? A comprehensive literature search was conducted using targeted keywords, including "embedding models", "open-source embedding models", "MTEB benchmark evaluation", "contextual sentence embeddings", "word embeddings", and "multimodal retrieval-augmented generation". Searches were performed across major academic and technical repositories, including but not limited to google scholar, IEEE Xplore, arXiv, and Hugging Face, yielding an initial set of 1,480 records. After deduplication (320 records removed) and title–abstract screening, 215 studies remained for full-text eligibility assessment. A subsequent quality appraisal prioritized methodological transparency, benchmark reproducibility, and clarity of reported metrics, resulting in 110 eligible studies. From this corpus, structured data were extracted to support comparative analysis, including (i) operational tradeoffs such as parameter count, embedding dimension, and inference latency; (ii) architectural techniques, including subword modeling, self-attention distillation, and Matryoshka learning; and (iii) application contexts such as healthcare, fraud detection, and retrieval-augmented generation. Following thematic consolidation and historical valida-

tion, the corpus was refined to 70 core papers, which form the empirical foundation of this survey. A comprehensive view can be seen in Figure 3

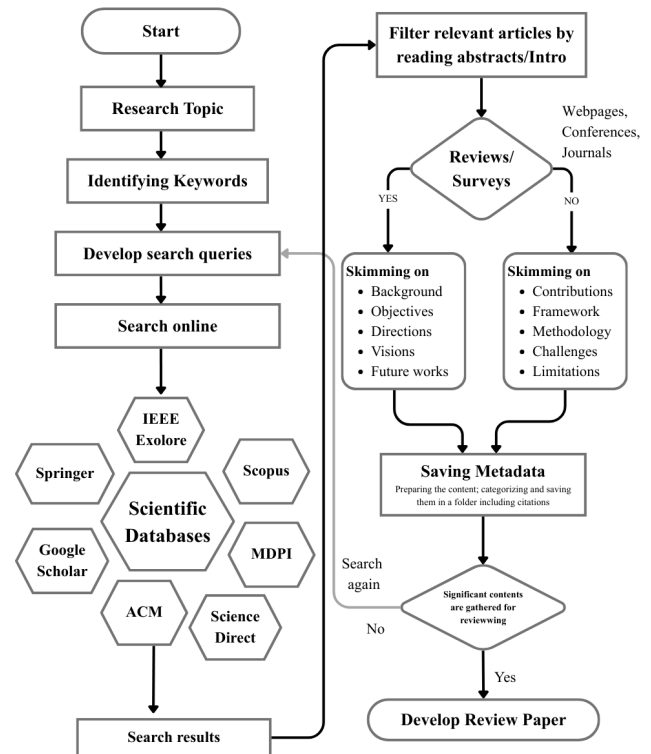


FIGURE 3: Systematic review workflow for the open source embedding models

D. CONTRIBUTIONS OF THIS PAPER

The primary contribution of this paper is to present a forward-looking and comprehensive examination of open-source embedding models, with emphasis on their performance characteristics and practical applications. Moving beyond a purely descriptive review of existing studies, this work adopts a deployment-focused and analytically grounded perspective on the open-source embedding landscape. Instead of simply listing available models, benchmarks, and use cases, we propose a structured framework that systematically links model architecture, benchmark performance, and real-world operational considerations. Specifically, we:

- 1) Trace the evolution of embedding models, from early word embeddings (Word2Vec, GloVe, FastText) to advanced contextual and multimodal approaches (BERT, Sentence-BERT, MPNet, MiniLM, BGE, GTE, and Nomic Embed).
- 2) Present a structured comparison of open-source embedding models, highlighting their strengths, weaknesses, and task suitability.

- 3) Review benchmarks such as MTEB and analyze their role in measuring embedding effectiveness, alongside limitations related to bias, dimensionality, and robustness.
- 4) Establish a decision-oriented selection framework that links model characteristics (e.g., contextual depth, multilingual capability, long-context support, efficiency mechanisms) to concrete application scenarios, enabling informed and resource-conscious model selection.
- 5) Explore practical applications across domains including semantic search, recommender systems, fraud detection, and healthcare.
- 6) Provide practical guidance for embedding model selection and deployment, balancing performance, interpretability, and resource constraints.

II. KEY OPEN-SOURCE EMBEDDING MODELS

The landscape of open-source embedding models is rich and diverse, offering a wide range of options for various NLP tasks. Among the foundational models are those focused on representing individual words in a vector space. This section presents a selection of common models and is not intended to be a comprehensive survey of all available open-source embedding models.

A. WORD EMBEING MODELS

1) Word2Vec

Word2Vec [4], a pioneering model developed by Google, learns word embeddings by analyzing the contexts in which words appear. It employs two primary architectures: the Continuous Bag of Words (CBOW), which predicts a target word from its surrounding context, and Skip-gram, which predicts the surrounding words given a target word. Its ability to capture semantic relationships by placing semantically similar words close together in the embedding space made it a highly influential model.

2) Continuous Bag of Words (CBOW)

The CBOW model predicts a target word w_t from its surrounding context words

$$w_{t-c}, \dots, w_{t-1}, w_{t+1}, \dots, w_{t+c} \quad (1)$$

where c is the context window size. The model aims to maximize the conditional probability $p(w_t | w_{t-c}, \dots, w_{t+c})$. Mathematically, this can be represented as:

$$p(w_t | w_{t-c}, \dots, w_{t+c}) = \frac{\exp(v'_{w_t} \top h)}{\sum_{w \in V} \exp(v'_w \top h)} \quad (2)$$

where v'_{w_t} is the output vector representation of the target word w_t , h is the context vector obtained by combining the input vectors of the context words (e.g., by averaging), and V is the vocabulary [32]

3) Skip-gram

In contrast, the Skip-gram model predicts the surrounding context words given the target word [33]. It aims to maximize the average log probability:

$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j} | w_t)$$

where T is the number of training words. The conditional probability $p(w_{t+j} | w_t)$ is often modeled using the softmax function, similar to CBOW:

$$p(w_{t+j} | w_t) = \frac{\exp(v'_{w_{t+j}} \top v_{w_t})}{\sum_{w \in V} \exp(v'_w \top v_{w_t})}$$

where v_{w_t} is the input vector of the target word and $v'_{w_{t+j}}$ is the output vector of a context word.

4) Global Vectors for Word Representation - GloVe

Another well-established word embedding technique is GloVe (Global Vectors for Word Representation), developed by researchers at Stanford University [19]. Unlike Word2Vec's local context focus, GloVe leverages the global co-occurrence statistics of words within a corpus to learn word embeddings. Its objective is to create a vector space where the dot product between the vectors of two words is related to the logarithm of their co-occurrence probability. This has proven to be particularly effective in capturing semantic relationships and performing well in tasks such as word analogy [34]. GloVe constructs a word-word co-occurrence matrix X , where X_{ij} represents the number of times word j appears in the context of word i . The model aims to learn word vectors v_i and v_j such that their dot product approximates the logarithm of the words' co-occurrence frequency.

The cost function J for GloVe is defined as:

$$J = \sum_{i=1}^V \sum_{j=1}^V f(X_{ij})(v_i \top v_j + b_i + b_j - \log X_{ij})^2 \quad (3)$$

where b_i and b_j are bias terms, and $f(X_{ij})$ is a weighting function that gives less weight to frequent word pairs. This function is designed to prevent the model from being dominated by very frequent words and to provide a cutoff for rare words. The relationship between the dot product and co-occurrence is key: GloVe learns a vector space where the dot product between word vectors reflects the logarithm of their co-occurrence probability, effectively capturing semantic relationships.

B. TEXT EMBEDDING MODELS

1) FastText

FastText, created by Facebook, extends the Word2Vec model by incorporating subword information [35]. Instead of treating words as atomic units, it breaks them down into character

n-grams and learns embeddings for these subword components. The embedding for a word is then represented as the sum of the embeddings of its constituent n-grams. This approach allows FastText to handle out-of-vocabulary words and capture morphological information, making it particularly advantageous for morphologically rich languages [36].

Moving beyond word-level representations, transformer-based models have revolutionized the field of embedding models.

2) Bidirectional Encoder Representations from Transformers BERT, developed by Google, is a prime example [37]. It generates contextualized word embeddings by analyzing the entire sequence of words in both left and right directions, leveraging the encoder segment of the transformer architecture and being pre-trained on large-scale text corpora using masked language modeling and next sentence prediction objectives [38]. This bi-directionality and the transformer's attention mechanism enable it to understand the context of words in a sentence more effectively than previous models.

The core of the transformer is the scaled dot-product attention, calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

where Q is the query, K is the key, V is the value, and d_k is the dimension of the keys. BERT uses multiple layers of these attention mechanisms, known as multi-head attention.

3) Sentence-BERT

While BERT excels at generating contextualized word embeddings, SBERT is a framework specifically designed to produce semantically meaningful embeddings for entire sentences. SBERT builds upon pre-trained BERT networks by adding a pooling layer to the output and then fine-tuning the model on sentence pair tasks using Siamese or triplet network structures [21]. This fine-tuning process optimizes the model to produce sentence embeddings that can be effectively compared using cosine similarity, making SBERT highly suitable for tasks like semantic search and text similarity.

4) Masked and Permuted Pre-training for Language Understanding

MPNet, developed by Microsoft Research Asia, represents a further advancement in pre-training techniques [12]. It combines the strengths of BERT's masked language modeling (MLM) and XLNet's permuted language modeling (PLM) while addressing their limitations [39]. In MLM, a certain percentage (e.g., 15%) of the input words are randomly masked, and the model's task is to predict the masked words based on the context. This forces the model to understand the context in both directions (left and right), leading to powerful contextualized representations. MPNet leverages the dependency among predicted tokens through permuted language modeling and takes auxiliary position information as input to reduce the discrepancy between pre-training and fine-tuning.

This innovative approach allows the model to achieve strong performance on a variety of language understanding tasks.

5) MiniLM

For applications requiring efficient embedding generation without sacrificing much in performance, MiniLM presents a compelling solution [40]. Developed through a process known as deep self-attention distillation, MiniLM is a distilled version of large transformer-based models like BERT [40], [41]. This technique trains a compact "student" model to replicate the self-attention behavior of the last transformer layer of a larger "teacher" model. The result is a model that is significantly smaller and faster, yet still retains much of the performance of its teacher.

6) BAAI General Embedding

Another powerful family of text embedding models is BGE (BAAI General Embedding), developed by the Beijing Academy of Artificial Intelligence [42]. They are known for their versatility and strong performance across various NLP tasks. BGE models—such as BGE-M3—support multiple retrieval paradigms, including dense, multi-vector, and sparse retrieval. They also offer impressive multilingual capabilities, supporting over 100 languages, and are capable of handling input sequences up to 8192 tokens, making them suitable for both short and long-form text.

7) General Text Embedding

Similarly, the GTE (General Text Embedding) models from Alibaba Group stand out for their performance in multilingual text embedding and retrieval [43]. Models like gte-multilingual-base excel in multilingual retrieval benchmarks, while others such as gte-Qwen2-7B-instruct and gte-large-en-v1.5 rank highly on the MTEB leaderboard. The latter is specifically optimized for English tasks and can process very long inputs, up to 8192 tokens.

8) Nomic Embed Models

Finally, Nomic Embed, developed by Nomic AI, offers efficient and high-performing models for both text and multimodal embeddings [44]. Nomic Embed Text achieves strong results compared to proprietary models like OpenAI's ada-002 and text-embedding-3-small, while maintaining a compact size and full reproducibility thanks to open-source training data and code. Meanwhile, Nomic Embed Vision extends these capabilities to joint text-image embeddings, enabling a range of multimodal applications.

C. MULTIMODAL EMBEDDING MODELS

Multimodal embedding models represent a shift from single-modality processing to a unified semantic space where different data types, such as text, images, and audio, are mapped to a shared coordinate system. This alignment allows the model to "understand" that the text "a golden retriever in the park" and an image of the same scene occupy the same semantic

neighborhood. The architectural foundation of multimodal embeddings has evolved from specialized, rigid structures to flexible, transformer-based unified encoders:

1) Contrastive Dual-Encoders:

This architecture uses two distinct backbones (e.g., a Vision Transformer for images and a BERT-based model for text). The models are trained using contrastive learning to pull "positive" pairs (matching image and text) together while pushing "negative" pairs apart.

2) Unified Multimodal LLMs:

A significant challenge in multimodal embeddings is handling varying input lengths and resolutions. **Nomic Embed Vision** [14] addresses this by providing high-performance, open-source embeddings that support long-context text and unified visual-textual search. By employing techniques like *Matryoshka Representation Learning (MRL)*, Nomic allows for flexible embedding dimensions, enabling users to truncate vectors for efficiency—a concept central to the "New Frontiers" of retrieval performance.

3) E5 and mmE5

The E5 (Emphasized Embedding) family introduced the concept of utilizing weakly supervised data followed by fine-tuning on high-quality synthetic data. The recent multimodal E5 **mmE5** [45] extends this to multimodality, using a "deep thinking" pass of a multimodal LLM to generate synthetic alignment data. This architectural approach treats the embedding task as a specialized distillation of a much larger multimodal LLM, resulting in state-of-the-art performance on the MMEB (Multimodal Embedding Benchmark).

4) CLIP (Contrastive Language-Image Pre-training)

CLIP [46] revolutionized the field by demonstrating that simple contrastive learning on a massive scale (400 million pairs) could create robust, zero-shot multimodal representations. It effectively proved that a dual-encoder architecture—consisting of a modified ResNet or ViT for vision and a Transformer for text—could learn highly transferable features without task-specific fine-tuning.

The progression from static, global embeddings (Word2Vec, GloVe) to dynamic, contextual representations (BERT, SBERT, MPNet) and finally to specialized, efficiency-focused models (MiniLM, BGE, GTE, Nomic Embed) marks the key evolutionary steps in the field. This diversity compels practitioners to carefully weigh model performance against deployment constraints. As summarized in Table 1, the decision necessitates a systematic analysis of the inherent Efficiency Tradeoffs, where achieving peak performance and long-context capabilities often correlates with higher latency and memory demands, while lightweight models like MiniLM and Nomic Embed offer optimized alternatives for resource-constrained environments. For example, the latency and memory descriptors in Table 1, are relative estimates

informed by common deployment observations rather than a single unified protocol. In practice, "Low" corresponds to inference times on the order of tens of milliseconds per embedding and memory footprints typically below ~ 1 GB (e.g., all-MiniLM-L6-v2 on CPU). "Moderate" indicates average latencies in the 100–300ms range with peak memory often around ~ 1 –2GB, while "High" refers to larger models with hundreds of milliseconds to over one second per query and memory demands exceeding ~ 2 –3GB on typical hardware. These bands reflect consistent patterns seen across community benchmarks for open-source embedding models.

III. TRAINING DATA AND DATASETS

A. COMMON CRAWL INITIATIVE

The Common Crawl initiative is a major open repository of web crawl data that has supported research and development across fields since its launch in 2008 [47]. It has archived petabytes of information, including raw web pages, metadata, and text extracts [48], offering a longitudinal view of the internet's evolving structure and content. Accessibility remains a core principle, with datasets stored on Amazon Web Services' Public Data Sets and available without requiring an AWS account. As of 2025, the dataset includes documents in many languages, though English dominates at about 60% [48], [49]. It also includes millions of PDFs and diverse web content, organized into three file types: WARC (full content), WAT (metadata), and WET (extracted text) [47], allowing flexible access based on research needs. Common Crawl data has broad applications, such as training natural language processing models, conducting computational linguistics research, performing market trend analyses, and studying cultural and political movements. Furthermore, the regular releases of host- and domain-level graphs provide deeper insights into the web's interconnected structure.

Nevertheless, the dataset has limitations. The automated crawl process tends to prioritize frequently linked domains, leading to an over-representation of popular websites and a lack of coverage for digitally marginalized communities. Language distribution is heavily skewed toward English, which may introduce biases into downstream tasks like word embedding training. Additionally, Common Crawl contains significant noise, including spam, offensive content, boilerplate text, duplicates, and outdated information. Due to its minimal curation approach, problematic content such as hate speech remains in the dataset, potentially reinforcing harmful biases when used for model training. While some filtered versions are available, filtering methods are often basic and do not eliminate deeper structural biases. Differences in processing pipelines can also leave distinct "fingerprints" on derivative datasets. Moreover, search-based access methods can introduce further selection bias, affecting the representativeness of extracted subsets.

B. WIKIPEDIA AND PROJECT GUTENBERG

Wikipedia, a widely popular online encyclopedia, has rapidly grown since its inception to encompass an extraordinarily

TABLE 1: Comparative overview of key open-source embedding models, highlighting architectural design, parameter scale, embedding dimensionality, and qualitative computational–efficiency tradeoffs.

Model	Primary Architecture	Model Size (Params)	Embedding Dimension	Computational Complexity	Efficiency (Latency/Memory)	Tradeoffs (Latency/Memory)
Word2Vec (Skip-gram/CBOW)	Shallow Neural Network	N/A (Algorithm)	50–300	Low	Extremely Low Latency; Minimal memory footprint; Fixed, non-contextual embedding.	
GloVe	Matrix Factorization	N/A (Algorithm)	50–300	Low	Extremely Low Latency for inference; Training requires high memory for co-occurrence matrix.	
All-MiniLM-L6-v2	Transformer (MiniLM)	22.7M	384	Moderate to High	Highest Transformer Efficiency; Very low memory (~80MB); Ideal for CPU/edge deployment.	
All-Mpnet-Base-v2	MPNet Encoder	109M	768	High	Efficiency Baseline: Excellent balance of accuracy and speed; Moderate memory footprint (~420MB).	
Nomic-Embed-Text-v1	Transformer (Custom)	137M	768	Moderate to High	Memory/Storage Optimized: Supports Matryoshka and Binary Quantization for significant memory reduction.	
Bge-Large-en-v1.5	BERT Encoder	335M	1024	High	Performance Focused: High MTEB score; Higher latency and memory consumption; Better suited for GPU inference.	
GTE-large-en-v1.5	Transformer++	434M	1024	High	Long Context Specialist: Highest computational demand in this group; Supports 8192 Max Tokens (increasing memory/latency for long inputs).	

large corpus with broad scope and constant updates. Available under the GNU Free Documentation License [50], it offers a vast amount of freely accessible text, making it a valuable resource for training word embeddings. The English version of Wikipedia alone contains millions of articles, and the platform supports over 300 different languages, making it ideal for the collection of sample texts for multilingual corpora [51], [52]. The articles within Wikipedia generally adhere to a consistent structure and are organized into category hierarchies, which can aid in learning semantic relationships within specific domains.

However, the collaborative editing nature of Wikipedia also presents challenges. The accuracy of its content is frequently questioned, and systemic bias means that quality and coverage can be uneven [53]. Variations in English dialects, for instance, can introduce differences in semantics, diction, and spelling. The sheer volume of text, exceeding a terabyte for the English version, necessitates efficient processing techniques [54]. While it strives for a neutral point of view (NPOV), political bias has been observed, with sentiment analysis suggesting a tendency to associate right-of-center public figures with more negative sentiment compared to left-of-center figures. Bias can also manifest in the portrayal of certain topics across different language versions, such as the negative emphasis on LGBT individuals in Russian Wikipedia compared to English or French versions [55]. Subjectivity and framing bias, where an author's perspective is presented as absolute truth, can also be present. The lack of diversity within the Wikipedia editor community, which

is predominantly white and male, is another factor that can contribute to biases in the content [56], [57]. Furthermore, the corpus contains noise in the form of contradictions, vandalism, and edits unrelated to bias mitigation.

Project Gutenberg on the other hand, is a long-standing volunteer effort focused on digitizing and archiving cultural works, primarily consisting of copyright-free literary works (books). Founded in 1971, it is the oldest digital library, with a collection exceeding 70,000 items. While primarily featuring literary fiction, it also includes non-fiction titles such as biographies, histories, and reference works. The majority of its releases are in the English language, but a significant number of non-English works are also available. The standard format for texts in Project Gutenberg is plain text, commonly using the Latin-1 character set, although some are in UTF-8 Unicode. This focus on plain text aims to ensure long-term readability and accessibility.

A challenge with Project Gutenberg is the inconsistent structure of the raw texts, which often contain boilerplate information about the creation of the corpus and copyright details at the beginning and end of the actual book content. This necessitates preprocessing to extract the main text. Furthermore, there is no single, universally agreed-upon full version of Project Gutenberg, and different studies often employ vastly different pre-processing strategies, raising concerns about the reproducibility of research results. Bias can also be present in the corpus, including gender and religious bias, reflecting the perspectives and societal norms of the authors and the time periods in which the works were written. Given

that many of the texts are digitized from older printed books, there is also the potential for errors introduced during the optical character recognition (OCR) process. To address some of these issues, the Standardized Project Gutenberg Corpus (SPGC) has been developed as an open science approach to a curated version of the complete PG data, aiming to provide a more consistent and reliable resource for statistical analysis of language. A summary of the data corpora can be seen in Table 2

IV. NEW FRONTIER IN MULTIMODAL EMBEDDING: EFFICIENCY AND UNIFICATION

The quest for computational efficiency in multimodal embeddings is no longer a secondary concern but a primary design principle. Recent architectural innovations are fundamentally changing how these representations are created and utilized. A core strategy is dimensionality reduction [58], which allows for the representation of high-dimensional data, such as images where each pixel is a dimension, into a lower-dimensional space. This process, by identifying commonalities and patterns, reduces the computational power and time required for deep-learning models to learn and infer from data.

A notable technique that embodies this principle is the Matryoshka embedding [59], [60]. Models like Voyage-3-large and Nomic Embed Vision utilize this approach, which is designed to concentrate the most important semantic information within the initial dimensions of the output vector. This allows the vector to be truncated to a smaller size without a significant loss of semantic information, thereby balancing high performance with lower inference costs and faster search times due to a reduced search space.

Another critical innovation addresses the data bottleneck. Multimodal Large Language Models (LLMs) are now being leveraged to generate high-quality synthetic datasets that cover a wide range of tasks and modality combinations [61], [62]. This addresses a significant historical challenge: the limited availability of labeled multimodal data, which has often hindered embedding model performance. The mmE5 model [45] is a prime example of this approach, achieving state-of-the-art performance on the MMEB Benchmark by training on a large-scale, multilingual dataset synthesized via a deep thinking process within a single pass of a multimodal LLM.

The evidence points to two distinct, yet complementary, approaches to enhancing multimodal embedding efficiency. The first, exemplified by mmE5, focuses on improving the quality of the training data to build a more robust and accurate traditional embedding model. This is a classical machine learning strategy where a superior input leads to a superior output. The second, which is more radical, re-architects the retrieval process itself to bypass the inherent scalability limitations of traditional embedding-based search. This intellectual divergence suggests that the future of multimodal systems may involve a bifurcated strategy, where different optimizations are applied depending on the specific

application's needs. One approach may be deployed for creating general-purpose embeddings, while another, more specialized one is used for high-speed, large-scale information retrieval.

V. COMPARATIVE ANALYSIS OF STRENGTHS AND WEAKNESSES

Each open-source embedding model discussed comes with its own strengths and trade-offs, making them suitable for different NLP tasks and application scenarios as can be seen in Table 3. For instance, the Word embedding models such as Word2Vec, GloVe, and FastText are widely recognized for their efficiency in processing large datasets and their ability to capture semantic relationships between words. These models generate embeddings based on word co-occurrence patterns, making them relatively fast to train, particularly in the case of Word2Vec and FastText. However, a key limitation of these models is that they produce static embeddings. This means a word is always represented by the same vector, regardless of the context in which it appears. For example, the word "bank" would have the same embedding whether it's used in the sentence "He went to the bank to withdraw cash" or "The river overflowed its bank." This lack of contextual awareness can reduce accuracy in tasks requiring nuanced understanding. Among these, FastText offers an edge by incorporating subword information (character n-grams), which allows it to handle out-of-vocabulary words and capture morphological variations. This is particularly useful for morphologically rich languages or when dealing with noisy, user-generated content such as social media text or OCR output. However, this added capability increases memory usage compared to Word2Vec and GloVe.

Transformer-based sentence embedding models, such as BERT, Sentence-BERT, MPNet, and MiniLM, have significantly advanced the ability to capture contextual meaning in natural language processing. BERT and MPNet exhibit strong performance across a wide array of NLP tasks, demonstrating their versatility. Sentence-BERT, specifically optimized for generating high-quality sentence embeddings, excels in tasks involving semantic similarity. In contrast, MiniLM, developed through knowledge distillation, strikes a balance between computational efficiency and performance. However, a common drawback of these models, particularly BERT and MPNet, is their substantial computational cost, which can be prohibitive in resource-constrained settings. Additionally, these models are often limited by the length of input sequences they can process. While pre-trained models offer robust general-purpose representations, fine-tuning on task-specific data is often necessary to achieve optimal results. These models are particularly well-suited for applications that require a deep understanding of context, including semantic search, text classification, question answering, and paraphrase detection.

In contrast, more advanced sentence and multimodal embedding models, such as BGE, GTE, and Nomic Embed, have set new benchmarks for performance. BGE models are

TABLE 2: Comparison of Common Text Sources for Word Embedding Training

Data Source	Size	Structure	Text Content	Noise/Bias	Suitability
Common Crawl	Petabytes, billions of pages monthly	WARC, WAT, WET files	Diverse web content	Noise, spam, hate speech, English-centric bias, prioritization of popular domains	High for general-purpose embeddings with careful filtering
Wikipedia	Terabytes, millions of articles	Structured articles with categories	Broad range of encyclopedic knowledge	Inaccuracies, systemic bias (political, cultural), vandalism	High for general and multilingual embeddings, but bias needs consideration
Project Gutenberg	Over 70,000 books	Plain text, HTML, etc.	Primarily literary and historical works	Boilerplate, inconsistent structure, potential OCR errors, historical language, gender/religious bias	Valuable for capturing language evolution and literary styles

particularly strong in multilingual contexts and support long-context inputs, making them ideal for processing extended documents. GTE models also excel in multilingual retrieval tasks, with certain variants optimized for English and capable of handling very long sequences. Nomic Embed, distinguished by its efficiency and reproducibility, offers a competitive edge while maintaining open-source transparency. The Nomic Embed Vision extends these capabilities to multimodal tasks, enabling joint embeddings of text and images. While BGE models have shown promising results, their generalizability to diverse real-world datasets remains under investigation, and GTE models may exhibit some performance inconsistencies across different languages. Additionally, some models within these families are still constrained by input length limitations. Nevertheless, these advanced models are particularly valuable for high-performance retrieval systems, multilingual applications, long-document processing, multimodal search, and any scenario where efficiency and transparency are critical.

VI. BENCHMARK RESULTS AND PERFORMANCE EVALUATION

The Massive Text Embedding Benchmark (MTEB) leaderboard serves as a central platform for evaluating and comparing the performance of various text embedding models across a wide range of natural language processing (NLP) tasks [30]. This platform offers a comprehensive overview of how different models perform on tasks such as retrieval, classification, clustering, and semantic similarity, among others. Models that rank highly on the MTEB leaderboard are often regarded as state-of-the-art in the field of text embeddings. For example, NV-Embed-v2 has achieved the top rank with an impressive performance across 56 different tasks, while gte-Qwen2-7B-instruct is also recognized as a top performer on this benchmark.

In terms of performance and computational efficiency, models like Sentence-BERT, including all-mpnet-base-v2 and all-MiniLM-L6-v2, offer a strong balance. While all-

mpnet-base-v2 generally provides better-quality embeddings, all-MiniLM-L6-v2 is notably faster, making it more suitable for applications that require low latency [63]. BGE models, particularly those developed by the BAAI, have demonstrated exceptional performance, especially on Chinese text embedding benchmarks, such as C-MTEB, and have gained considerable popularity within the NLP community. Nomic Embed also stands out, showing remarkable capabilities by even outperforming larger models in arena-style evaluations, which prioritize user preferences in retrieval scenarios.

When examining performance across specific NLP tasks, certain models excel in particular domains. For text classification tasks, BERT and its distilled variants like MiniLM, along with Sentence-BERT, consistently yield strong results [64]. FastText is another popular choice for text classification, particularly when dealing with large datasets, due to its speed and efficiency. In semantic similarity tasks, models like Sentence-BERT, including all-mpnet-base-v2 and all-MiniLM-L6-v2, are specifically trained to perform well, often achieving high scores on Semantic Textual Similarity (STS) benchmarks. Other models like MPNet and the BGE family also demonstrate excellent capabilities in capturing semantic similarity. For information retrieval tasks, models explicitly optimized for this purpose, such as BGE (particularly BGE-M3), GTE models, and Nomic Embed, frequently secure top rankings on the MTEB leaderboard. These models are specifically engineered to produce embeddings that accurately capture the relevance between search queries and documents.

The performance of these embedding models on various benchmarks is influenced by several factors, including the training data, model size, and architecture. Models trained on larger, more diverse datasets tend to achieve higher scores, as they are exposed to a wider range of linguistic patterns and semantic relationships [30]. Model size, often correlated with performance, is typically measured by the number of parameters, with larger models having a greater capacity to

TABLE 3: Comparison of Embedding Models: Strength and Weaknesses

Model	Type	Strengths	Weaknesses	Ideal Use Cases
Word-Level Models				
Word2Vec	Word	Fast, captures semantics	Static, poor with polysemy	Baseline similarity, low-resource settings
GloVe	Word	Global statistics, analogies	Static, weak on small corpora	Analogy tasks, baseline embeddings
FastText	Word	Handles OOV, morphology	Higher memory use	Rich morphology, unseen words
Contextual Models				
BERT Sentence-BERT	Contextual Word Sentence	Strong bidirectional context Optimized for similarity	Expensive, fixed length Inherits BERT limits	Classification, QA, NER Semantic search, paraphrase detection
MPNet	Sentence	Models dependencies well	Expensive, fixed length	Language understanding tasks
MiniLM	Sentence	Lightweight, efficient	Slightly lower accuracy	Low-latency, resource-constrained
BGE Models	Sent./Multilingual	SOTA performance, long context	Variable generalizability	Multilingual retrieval, long documents
GTE Models	Sent./Multilingual	Efficient, long sequences	Language variance	Multilingual retrieval, long context
Nomic Embed	Contextual/ Multimodal	High perf., reproducible, multimodal	Dataset-specific eval limits	Efficient retrieval, multimodal applications

learn complex patterns [65], [66]. However, larger models also require significantly more computational resources for both training and inference. The underlying architecture also plays a crucial role in performance. Transformer-based architectures have generally outperformed earlier recurrent and convolutional neural network-based models in many NLP tasks, due to their ability to capture long-range dependencies and contextual information effectively [30], [67]. Furthermore, fine-tuning pre-trained models on task-specific data can significantly enhance performance for particular downstream applications [68]. Table 4 below summarizes the performance of several top models on the MTEB leaderboard, providing insight into their relative rankings across various tasks such as retrieval and semantic textual similarity (STS), as well as their model size and embedding dimensions.

Entries in Table 4 mix approximate values and missing cells because the underlying MTEB leaderboard and community benchmarks do not always report all subtask scores or exact model footprints in a consistent format. Some models (e.g., NV-Embed-v2, bge-large-en) have complete averaged retrieval and STS scores available from public leaderboard sources, while others (such as gte-Qwen2-7B-instruct and nomic-embed-text-v1) may not have distinct retrieval or STS breakdowns published or may aggregate results differently on external leaderboards. Likewise, model size entries are often approximated based on published parameter counts or commonly observed memory footprints (e.g., quantized sizes reported on model hubs) rather than a single standardized measurement. These approximations and omissions reflect the current state of publicly available benchmark data and variations in reporting rather than omissions in the evaluation methodology.

A. NOVEL EVALUATION METRICS FOR A NUANCED ASSESSMENT

The reliance on established benchmarks like the MTEB is no longer sufficient for a comprehensive evaluation of modern embedding models. MTEB, while foundational, is limited by its scope, which lacks diversity in languages, domains, and tasks. A new evaluation paradigm is required to address the expanded capabilities of modern models.

To meet this need, the Massive Multilingual Text Embedding Benchmark (MMTEB) has emerged as a large-scale, community-driven expansion of MTEB. MMTEB covers over 500 quality-controlled evaluation tasks across more than 250 languages. The benchmark's tasks are diverse and challenging, including instruction following, long-document retrieval, and code retrieval, representing the largest multilingual collection of evaluation tasks to date. To facilitate accessibility and reduce computational costs, MMTEB employs a novel downsampling method based on inter-task correlation, ensuring a diverse selection of tasks while preserving the relative model rankings.

The introduction of MMTEB signals a significant shift in the research agenda. While MTEB sought to identify "universal text embeddings" that perform well across a broad range of tasks, MMTEB's emphasis on specific, challenging tasks suggests a move towards evaluating specialized capabilities. This implies that future models may be highly specialized for particular tasks and domains rather than being a single, one-size-fits-all solution. This change in focus demonstrates the field's increasing maturity and its progression from simple generalization to a more nuanced understanding of specialized performance.

TABLE 4: MTEB Benchmark Results and Model Information

Model	Average Score	MTEB	Retrieval age	Average	STS Average	Model Size	Embedding Dimension
NV-Embed-v2 [69]	72.31		62.65		84.31	~60KB (config) / ~7.85GB (parameters)	4096
Gte-Qwen2-7B-instruct [43]	~70.24		Not Reported		Not Reported	~7B parameters / 8.1GB (Q8_0 quantized)	3584
Bge-Large-en-v1.5 [70]	64.23		60.03		83.11	1.34GB	1024
All-Mpnet-Base-v2 [12]	63.30		57.02		69.57	420MB	768
All-MiniLM-L6-v2 [41]	58.80		49.54		68.06	80MB	384
Nomic-Embed-Text-v1 [44]	62.39		Not Reported		Not Reported	0.55GB	768

B. LONG-CONTEXT CAPABILITIES: NEW BENCHMARKS AND CHALLENGES

The ability of models to process and understand long contexts is a crucial capability for real-world applications such as legal review and financial analysis. Early benchmarks, such as "Needle-in-a-Haystack," [71] provided only a "surface form of long-context understanding" by simply embedding a key phrase within irrelevant noise. This simplification does not reflect the complex, multi-document reasoning required in real-world scenarios.

To address this, new and more sophisticated benchmarks have been developed. Datasets like L-Eval [72] and LongBench [73] offer a significant step forward, covering diverse fields from law to economics and including both English and Chinese texts. However, the most recent and advanced benchmark is Loong [74], which is specifically designed for extended multi-document question answering. The core principle of Loong is to "leave no document behind," meaning every document in the context is relevant to the final answer, forcing the model to process the entire context and synthesize information from multiple sources. Loong includes four novel task types—Spotlight Locating, Comparison, Clustering, and Chain of Reasoning, that evaluate different facets of multi-document understanding.

The progression from LongBench to Loong signifies a critical shift from simple information retrieval to multi-document reasoning. The former primarily tests a model's ability to recall information from a long document, while the latter, with its "Chain of Reasoning" task, assesses the model's capacity to perform multi-step, cross-document inference. This mirrors the trend in the broader LLM field toward massive context windows and demonstrates that evaluation methodologies are evolving to keep pace with the increasingly complex cognitive tasks that modern models are capable of performing.

As illustrated in Table 5, modern open-source models demonstrate a clear correlation between parameter scale and their proficiency in complex reasoning tasks. While smaller models like *all-MiniLM-L6-v2* remain efficient for short-form semantic similarity, they lack the architectural capacity for the "Chain of Reasoning" and "Spotlight Locating" tasks

central to the Loong benchmark. Performance is primarily quantified via the nDCG@10 (Normalized Discounted Cumulative Gain at rank 10) metric. This metric is critical for long-context evaluation as it rewards models for placing highly relevant documents at the top of the retrieval list, applying a logarithmic discount to relevant items appearing at lower ranks to reflect the limited processing window of downstream LLMs.

C. MEASURING COMPUTATIONAL AND INFERENTIAL EFFICIENCY

The definition of "efficiency" in the context of embedding models has expanded beyond simple metrics like FLOPs (Floating-Point Operations Per Second) or training time. With the shift to LLMs for embedding tasks, inference costs have become a "persistent concern". To address this, researchers are exploring various strategies to reduce the size and complexity of models while maintaining performance. These include pruning, which removes unnecessary parameters; quantization, which reduces numerical precision (e.g., from 32-bit to 8-bit); and knowledge distillation, which trains a smaller model to mimic a larger one.

Latency and model size are also critical metrics. Models like Voyage-3-lite exemplify a pragmatic approach to deployment, achieving a strong cost-to-performance ratio with a smaller output size of 512 dimensions. This architectural choice leads to proportionally faster searches compared to models with larger dimensions, such as Voyage-3-large (2048 dimensions). The mmE5 model further illustrates this evolving definition of efficiency by achieving state-of-the-art results with a comparatively modest 560 million parameters, a stark contrast to the billions of parameters in other LLMs. This reflects a field that is moving away from the "bigger is better" mindset towards a more practical, application-driven approach where efficiency is a primary design constraint for real-world usability.

D. LIMITATIONS OF BENCHMARK EVALUATION

While benchmarks like MTEB provide a valuable tool for comparing and evaluating the performance of embedding models, it's crucial to acknowledge their limitations. Re-

TABLE 5: Performance of Open-Source Embedding Models on Long-Context Reasoning Benchmarks

Model	Context Window	nDCG@10	Loong Specialization	Params
Qwen3-Embed-8B	32,768	68.2	Chain of Reasoning, Comparison	8B
NV-Embed-v2	32,768	66.8	Spotlight Locating, Retrieval	7.8B
gte-Qwen2-7B-instruct	32,768	64.4	Multi-document Synthesis	7B
Jina-embeddings-v3	8,192	60.1	Task-Specific (Retrieval/Clustering)	570M
BGE-M3	8,192	58.9	Multi-lingual, Clustering	567M
nomic-embed-v1	8,192	49.2	Global Context Mapping	137M
all-MiniLM-L6-v2	512	< 35.0	Short-form Similarity	23M

lying solely on benchmark results can sometimes lead to a skewed perception of a model's true capabilities and its suitability for real-world applications. One key limitation is that benchmarks may not cover all relevant downstream tasks [75]. Benchmarks are designed to assess performance across a range of common NLP tasks, but they might not include the specific tasks or nuances of every application [76]. A model that performs well on a general benchmark might not necessarily excel in a highly specialized or domain-specific task. However, newer benchmarks, such as BEIR [77] and those based on SuperGLUE principles [78], are actively being developed to mitigate this overfitting by incorporating more diverse and challenging evaluation scenarios. These benchmarks emphasize robustness and generalization, offering a more reliable assessment of embedding model quality.

Furthermore, benchmarks can be susceptible to overfitting [79]. Developers might inadvertently optimize their models to perform well on specific benchmark datasets, potentially sacrificing generalization ability [80]. This can result in models that achieve high scores on benchmarks but fail to perform as effectively when applied to unseen data or real-world scenarios. Evaluation metrics used in benchmarks may also not fully capture the nuances of semantic understanding. While metrics like accuracy, F1-score, and cosine similarity provide quantitative measures of performance, they might not always reflect the subtle aspects of language comprehension that are crucial for many NLP applications [81].

It is also important to consider the potential biases present in benchmark datasets [82], [83]. If the datasets used for evaluation contain inherent biases related to gender, race, or other sensitive attributes, the benchmark results might not accurately reflect a model's fairness or its potential for generating biased outcomes. Addressing these limitations requires a more holistic approach to evaluation, combining benchmark results with thorough testing on real-world data and careful consideration of ethical implications.

VII. CHALLENGES IN USING EMBEDDING MODELS

While embedding models have proven to be incredibly powerful tools in NLP, their application is not without its challenges and potential pitfalls. One significant concern is the issue of bias [83]. Embedding models are trained on vast amounts of text data, and if this data contains societal biases related to gender, race, or other demographic attributes, these biases can be inadvertently encoded into the vector

representations learned by the model [82]. This can lead to skewed or unfair outcomes in downstream applications, such as search results that disproportionately favor certain groups or recommendation systems that reinforce existing stereotypes. Addressing bias in embedding models is therefore a critical ethical consideration, requiring careful data curation and the development of bias mitigation techniques. Several strategies can be employed to detect and mitigate bias. Detection methods include analyzing embeddings for skewed representations of gender, race, or other sensitive attributes using metrics like the Word Embedding Association Test (WEAT) or similar bias measurement techniques [84]. Mitigation strategies can be broadly categorized into: Pre-processing techniques, which involve modifying the training data to balance representation or removing biased examples; In-processing techniques, which modify the training objective or model architecture to reduce bias during training, such as adversarial debiasing; and post-processing techniques, which adjust the learned embeddings after training to remove biased information, like geometric debiasing.

Another challenge arises from the dimensionality of the embeddings produced by these models. Modern embedding models often generate vectors with hundreds or even thousands of dimensions (e.g., 768, 1024, or 1536 dimensions are common). While higher dimensionality can enable the capture of richer semantic information, it also leads to increased storage requirements and computational costs [85]. Furthermore, very high-dimensional data can sometimes suffer from the "curse of dimensionality," where the performance of certain machine learning algorithms degrades as the number of features (dimensions) increases [86]. Techniques for dimensionality reduction, such as Principal Component Analysis (PCA) or Matryoshka Representation Learning (MRL) [60], [87], can be employed to mitigate these issues by reducing the number of dimensions while preserving most of the essential information. For instance, the Figure 4 shows how PCA is leveraged to reduce the complexity of text representations generated by nomic embed model using 14 sample sentences, some IT related an others not.

IT-Related: Python programming, artificial intelligence and machine learning, cloud computing (AWS, Azure), PostgreSQL databases, cybersecurity, web development (React, Node.js), computer networking (TCP/IP).

Semi-Related: Online education platforms (LMS), digital marketing, electronic music production (DAWs).

Unrelated: Baking sourdough bread, classical music con-

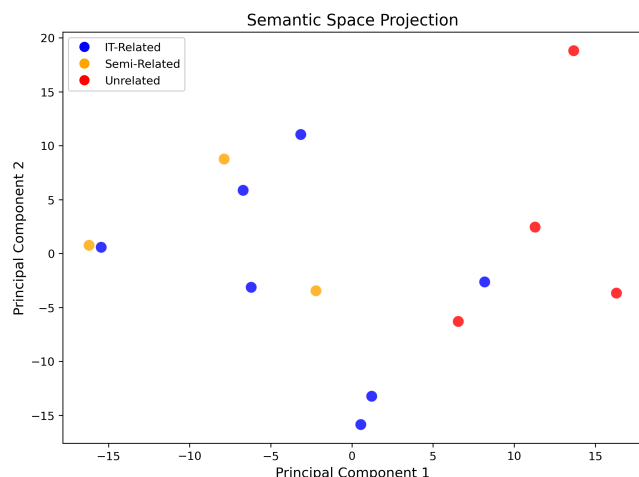


FIGURE 4: PCA Scatter Plot of Mixed Items using Nomic Embed Text Model to show the reduction of text representation from many dimensions to 2D

certs, gardening techniques, ancient Roman architecture.

These representations, which capture the semantic meaning of phrases, typically exist in a very high-dimensional space. By applying principal component analysis, these high-dimensional data is projected onto a two-dimensional plane. This allows for visual inspection of how different text items are positioned relative to each other. The expectation is that items with similar meanings or belonging to related categories will be located closer together on the resulting scatter plot, while dissimilar or unrelated items will be further apart. This visual clustering provides an intuitive way to assess how well the language model's embeddings capture semantic relationships and distinctions between various concepts.

By applying the PCA, the high-dimensional embeddings are projected onto the two-dimensional plane shown in Figure 4. This visualization confirms the semantic integrity of the Nomic Embed model's representations. Specifically, the IT-Related Concepts form a tight, highly cohesive cluster, demonstrating the model accurately perceives their strong semantic overlap. Conversely, the Unrelated Concepts (e.g., 'Baking sourdough bread', 'Ancient Roman architecture') are widely dispersed and positioned at significant distances from the central IT cluster, highlighting the model's ability to differentiate dissimilar topics. The Semi-Related Concepts (e.g., 'Digital marketing') occupy an intermediate space, serving as a semantic bridge. The practical significance of this result is that it validates the semantic structure is well-preserved even after dimensionality reduction, directly supporting the rationale for using MRL and other dimensionality reduction techniques in deployment.

Embedding models can also exhibit sensitivity to input variations. Even minor alterations in the input text, such as typos, grammatical errors, or changes in phrasing, can sometimes result in significantly different embeddings [88]. This sensitivity can negatively impact the performance of down-

stream tasks, especially in real-world applications where user input might be noisy or inconsistent [89]. Pre-processing techniques like spell checking, grammar correction, and text normalization can help to improve the robustness of embedding models to such variations. Additionally, models like FastText, which utilize subword embeddings, inherently show some resilience to out-of-vocabulary words and misspellings [90]. Finally, contextual limitations can pose a challenge, particularly when dealing with long documents or extended conversations. While transformer-based models like BERT provide contextual embeddings, they often have a fixed context window, limiting the amount of surrounding text they can consider when generating an embedding (e.g., 512 tokens for many BERT-based models). This can be problematic when the relevant context extends beyond this window. Newer models like BGE-M3 and gte-large-en-v1.5 offer larger context windows (up to 8192 tokens), which can help in capturing broader contextual dependencies. For models with smaller context windows, techniques like document chunking and using embeddings from multiple chunks can be employed to address this limitation.

VIII. FINETUNING STRATEGIES

Embedding models traditionally assume static environments; however, real-world systems, particularly in robotics and knowledge-intensive applications, often operate in dynamic settings where new entities and relations continuously emerge. This discrepancy introduces significant challenges, notably the need for frequent retraining and the phenomenon of catastrophic forgetting, where models lose previously acquired knowledge when exposed to new data. To address these limitations, recent research has explored continual learning and efficient finetuning strategies that enable embeddings to adapt incrementally without full retraining, while also investigating causal approaches to enhance interpretability and robustness.

A. CONTINUAL LEARNING FOR EMBEDDINGS

Traditional knowledge graph embedding (KGE) models such as TransE and Analogy operate under the assumption of a fixed set of entities and relations, making them ill-suited for dynamic environments. In scenarios where new facts are continuously observed, these models require complete retraining, resulting in high computational overhead and vulnerability to catastrophic forgetting [91]. To overcome these challenges, Daruna et al. introduce the Continual Knowledge Graph Embedding (CKGE) framework, which integrates continual learning principles into KGE models. CKGE enables incremental updates of embeddings as new knowledge becomes available, eliminating the need to retain all historical data and significantly reducing retraining costs.

One of the central methods explored in CKGE is Deep Generative Replay (DGR), a technique designed to maintain stability across learning sessions. As illustrated in Figure 5, the DGR architecture is based on a Variational Autoencoder (VAE) framework, where input triples representing head,

relation, and tail entities are embedded and processed through a Gated Recurrent Unit (GRU)-based encoder. This encoder projects the input into a latent space that captures semantic patterns in a compressed form. A corresponding decoder, composed of GRU layers and a softmax output, reconstructs prior knowledge from the latent representation, enabling the system to "replay" previously learned facts without explicitly storing all past data. Linear transformation layers, indicated by white outlines in the architecture, facilitate transitions between embedding spaces, latent variables, and predictions. Through empirical evaluations on benchmarks such as WN18RR and FB15K237, Daruna et al. demonstrated that DGR significantly mitigates forgetting, albeit at the cost of increased training time compared to simpler regularization-based methods.

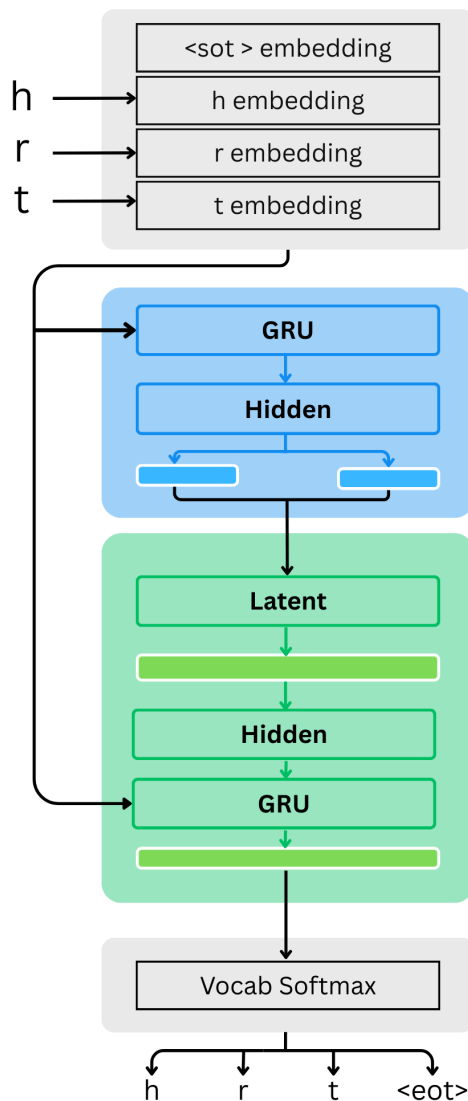


FIGURE 5: DGR architecture. Layers with white outline are linear [91]

B. EFFICIENT METHODS FOR UPDATING EMBEDDINGS

Beyond DGR, several efficient techniques have been proposed to update embeddings in dynamic environments without exhaustive retraining. Regularization-based approaches such as L2 regularization and Synaptic Intelligence selectively constrain parameter updates, penalizing deviations from previously important weights and thereby preserving earlier knowledge [91]. Architectural innovations like Progressive Neural Networks and Copy Weight Re-initialization further enhance continual learning by allocating separate network components for new tasks or by strategically resetting weights to integrate new information. These methods are particularly valuable in resource-constrained settings, where storage, computation, and time are limited. Collectively, they make incremental embedding updates practical and scalable, supporting real-time applications in fields like robotics, recommendation systems, and edge computing.

C. CAUSAL EMBEDDINGS FOR INTERPRETABILITY AND ROBUSTNESS

While continual learning addresses the dynamics of evolving knowledge, another emerging frontier focuses on embedding causal reasoning directly into model architectures. Causal embeddings aim to move beyond the predictive capabilities of traditional models by explicitly modeling causal relationships, thus enabling interventions and counterfactual reasoning. Unlike standard deep learning models that act as black boxes, causal embeddings provide greater transparency and fairness by answering "what-if" scenarios based on learned causal structures [92]. Li et al. introduced causal embedding frameworks tailored for recommendation systems. In their work on "Multimodal Graph Causal Embedding for Multimedia-based Recommendation," they leverage multimodal data sources to model causal dependencies between entities, improving both recommendation accuracy and system interpretability. Similarly, their CausalCDR framework for cross-domain recommendation embeds causal knowledge to facilitate the transfer of insights across domains while maintaining fairness and reliability. The architecture of causal embeddings typically involves processing raw input features through encoders that extract high-level causal concepts, which are then used for outcome prediction and counterfactual analysis. By embedding causal reasoning into the learning process, models become more robust against biases and more transparent to end users, especially in sensitive domains such as healthcare, finance, and personalized recommendations.

IX. APPLICATIONS OF EMBEDDING MODELS IN VARIOUS DOMAINS

Embedding models have found widespread applications across a multitude of domains, demonstrating their versatility and power in understanding and processing natural language. In search engines, embedding models enable semantic search capabilities, allowing users to find relevant information even when their queries use different keywords that have the same

underlying meaning [93], [94]. By matching the semantic meaning of a query with the semantic meaning of documents, search engines can provide more accurate and comprehensive results. Recommendation systems also heavily rely on embedding models to understand user preferences [95], [96]. By representing user interests and item characteristics as embeddings, these systems can identify semantic similarities and provide more personalized and relevant recommendations. In the realm of security, fraud detection can benefit from embedding models by representing user behavior or transaction patterns as vectors [97]. Anomalous patterns in the embedding space can then be identified as potential indicators of fraudulent activity [98]. The healthcare domain has also seen increasing adoption of embedding models for tasks like medical diagnosis [99], [100]. By applying these models to medical texts such as patient records, research papers, and clinical notes, it becomes possible to identify similar cases, extract crucial information, and potentially assist in the diagnostic process. Specialized models like BioBERT and ClinicalBERT, which are BERT variants trained on biomedical text, are particularly valuable in this field [101], [102]. Information retrieval is a critical area where embedding models significantly contribute, particularly in the context of Retrieval-Augmented Generation (RAG) systems [103], [104]. In RAG systems, embedding models are used to retrieve relevant documents or passages from a large corpus based on a user's query. The retrieved information is then integrated into the input of large language models, enhancing their ability to generate more accurate, contextual, and informative responses. This retrieval capability is also essential for clustering and classification tasks, where embedding models group semantically similar data points, enabling efficient content organization. These tasks support applications such as sentiment analysis, topic modeling, and document categorization [105]. While machine translation is not the primary focus for most embedding models, they play a crucial role in representing the meaning of text in different languages, with multilingual embedding models being essential for cross-lingual tasks.

The practical applications of open-source embedding models can be demonstrated through several examples across various domains. For instance, in semantic search, models like Sentence-BERT (e.g., all-mpnet-base-v2) are used to generate embeddings for documents within a knowledge base. When a user submits a query, the model compares the query's embedding to those of the documents, retrieving the most semantically relevant articles. In recommendation systems, models like Word2Vec or FastText learn embeddings for products based on user purchase history. By measuring the similarity between the embeddings of previously purchased items and other products, the system can suggest new items that align with the user's preferences. In the case of sentiment analysis, a BERT model can be fine-tuned on a labeled dataset of customer reviews to predict the sentiment of unseen reviews. Similarly, in question answering systems utilizing RAG, high-performance models like MPNet or BGE

can retrieve relevant context from a document collection, providing the necessary information for a large language model to generate an accurate response. Lastly, in the medical domain, models like BioBERT can be applied to patient notes to find similar cases, aiding in diagnosing conditions and planning treatments by identifying patterns from related patient data.

A. THE SHIFT TO GENERATIVE RETRIEVAL

The most profound re-engineering of the retrieval process is the emergence of generative retrieval [106], [107]. This cutting-edge approach, showcased by models like GENIUS, represents a paradigm shift away from traditional vector search. Conventional embedding-based methods map queries and database items into a high-dimensional space and then perform a nearest-neighbor search to find the most similar vectors. This approach, however, faces significant scalability challenges, as the cost of building and maintaining indexes grows with the size of the dataset, leading to slower retrieval times for large collections.

In contrast, generative retrieval, as demonstrated by the GENIUS model presented at the 2025 Conference on Computer Vision and Pattern Recognition (CVPR) [108], eliminates the need for expensive index building. Instead, the generative model directly outputs the unique identifier (ID) of the target data item. This enables constant-time retrieval, meaning the speed of the search remains constant regardless of the database size. The model achieves this through two core innovations: semantic quantization, which creates a sequence of codes as a target ID, and query augmentation, which helps the model generalize to new data by interpolating between initial queries and target IDs.

This transformation challenges the very foundation of current Retrieval-Augmented Generation (RAG) systems. The generative model is no longer a simple vectorizer; it has become the entire retrieval engine. This means that future multimodal systems may move beyond traditional vector databases, relying instead on highly efficient generative models that can produce direct, quantifiable search results. This represents a significant step forward in optimizing both the accuracy and efficiency of multimodal information retrieval.

X. LIMITATIONS

This survey provides a structured, deployment-aware perspective on open-source embedding models while operating within several defined boundaries. Rather than establishing definitive rankings, it treats benchmark results as time-dependent snapshots, acknowledging that performance is highly sensitive to evaluation protocols and domain-specific characteristics. The analysis of efficiency, scalability, and operational trade-offs is derived from published reports and documentation rather than a unified, controlled experimental evaluation, meaning that variables such as hardware configurations, numerical precision, and batch settings may limit strict comparability. Furthermore, the scope is constrained by the inconsistency of technical reporting across the lit-

erature, particularly regarding training data, "open-source" licensing nuances, and the emerging field of multimodal representations which can impact traceability. Ultimately, the work discusses practical deployment, bias, and robustness at a conceptual and synthesis-driven level rather than through primary empirical validation, highlighting ongoing research challenges in standardizing evaluations across heterogeneous tasks and maintaining representation quality under domain shift.

XI. CONCLUSION AND RECOMMENDATIONS

This study has explored the rich landscape of open-source embedding models, highlighting their definition, significance, diverse types, and the crucial role of benchmarks in their evaluation. A comprehensive list of prominent open-source models, ranging from foundational word embeddings like Word2Vec and GloVe to advanced contextual and multimodal embeddings such as BERT, Sentence-BERT, MPNet, MiniLM, BGE, GTE, and Nomic Embed, has been presented. Each model exhibits unique strengths and weaknesses, making them suitable for different NLP tasks and application scenarios. Benchmark results from platforms like MTEB provide valuable insights into their relative performance on tasks such as text classification, semantic similarity, and information retrieval, underscoring the influence of factors like training data, model size, and architecture on their effectiveness. Common pitfalls and challenges associated with using embedding models, including bias, dimensionality issues, sensitivity to input variations, and contextual limitations, necessitate careful consideration and the application of appropriate mitigation strategies. The wide-ranging applications of embedding models across domains like search engines, recommendation systems, fraud detection, medical diagnosis, information retrieval, and content organization demonstrate their transformative potential in various industries. Furthermore, the ability of embedding models to enhance documentation through semantic search, content organization, improved linking, and personalized experiences highlights their value in creating more effective and user-friendly information resources.

Beyond benchmark performance, an important insight from this survey is that embedding model selection should be treated as an engineering decision rather than a purely accuracy-driven choice. Marginal improvements in benchmark scores often come at a disproportionate cost in latency, memory footprint, and operational complexity, particularly for large transformer-based models. Therefore, it is required to make a reasonable tradeoff between performance metrics, memory footprint, and operational complexity, and energy consumption to ensure acceptable performance, lower cost and to satisfy sustainability requirements.

Selecting the most suitable open-source embedding model for a specific need requires careful consideration of several factors. The primary driver should be the specific NLP task at hand, as different models are optimized for different applications. For tasks requiring fine-grained semantic understand-

ing at the word level, traditional word embeddings might suffice, while tasks involving contextual understanding of sentences or documents generally benefit from transformer-based models. The need for multilingual support is another critical factor, with models like BGE and GTE offering strong capabilities in this area. Computational efficiency and resource constraints should also be considered, as larger, more powerful models often come with higher computational costs. The length of the input text can also influence model selection, with newer models offering larger context windows for handling longer documents. Ultimately, the desired level of accuracy and the specific requirements of the application will guide the choice.

In practice, efficient sentence-level embeddings such as MiniLM-, MPNet-, or Nomic-based models frequently provide superior cost–performance tradeoffs for production systems, including retrieval-augmented generation and semantic search pipelines. High-capacity models are best reserved for accuracy-critical scenarios where latency and memory constraints are secondary. It is often beneficial to experiment with a few different models and evaluate their performance on a relevant dataset. In many cases, fine-tuning a pre-trained model on domain-specific data can lead to significant improvements in performance, making this an important step in optimizing the use of embedding models for particular applications.

Finally, this survey highlights the need for future research to move beyond accuracy-centric evaluation by incorporating standardized efficiency-aware benchmarks, broader multilingual and multimodal assessments, and adaptive embedding strategies that explicitly balance precision and computational cost at inference time. More fundamentally, the field lacks standardized, hardware-aware evaluation protocols that jointly measure accuracy, robustness, bias, energy consumption, and latency under realistic deployment conditions. Addressing these gaps, along with improving multilingual equity, multimodal consistency, and robustness under domain shift—remains a central research priority for the next generation of embedding systems.

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